

PhotoNavigator 2.0

Features & Benefits Exhibit

Comprehensive inventory of shipped capabilities
Version 2.0-beta.100 · July 13, 2026

Evidence: source tree, menus, SQL migrations, release notes, and in-app help.

1. Executive summary

PhotoNavigator is a Windows desktop **multi-panel light table** and digital asset workstation. It replaces the legacy wxWidgets edition while preserving **JSON session compatibility** (state.json, settings.json, named profiles such as party.json).

The product targets **power reviewers** — event photographers, real-estate inspectors, research labs, and content curators — who need to see many images at once, page hands-free, edit metadata in place, and share without round-tripping through email attachments alone.

Core differentiators: configurable 2–36 panel grids, signature **AutoBrowse** timed paging, 200-effect slideshow, optional **PostgreSQL catalog** (DAM), local and cloud **AI Photo Chat**, **Family Safety** with content shields, first-party **Photo Lab** computational editing, comprehensive **LibRaw camera RAW**, smooth free rotation via **PhotoNavigator Kernel**, and **PhotoNavigatorGlobalCommunity** — peer photo messaging, live multicam video, rights-first capsules, market listings, and pen-pal pairing.

Current release: 2.0-beta.100 · 775 automated tests · signed installer via GitHub Releases.

2. Light table & viewing

PhotoNavigator's primary surface is a **grid of panels**, each showing one image with independent zoom, pan, and rotation. Layout presets control how many panels appear per page.

- **Multi-panel grid:** 2–36 panels per page via toolbar layout slider; auto-fit letterboxing; customizable padding, gaps, pane borders, selection frame, and pop-in page animations.
- **Single-panel focus:** double-click any panel, thumbnail, or slideshow entry for full-resolution view.
- **TriView:** directory tree + thumbnails on the left; paged grid on the right.
- **ScrollView:** vertical scroll grid (e.g. 9×4 at layout 36) with smooth AutoBrowse scrolling.
- **Split Screen:** left/right or top/bottom dual-pane layouts.
- **Thumbnail strip:** entire active library; scrolls independently during AutoBrowse.
- **Loupe & Minimap:** hover magnifier and panel navigation aids.
- **Free rotation (beta.100):** middle mouse held + wheel — continuous 0.30° steps; GPU hyper-sample preview; outward matte hides residual zigzag (display-only).
- **Zoom:** wheel zoom without middle pressure; floor 5% of native size (was 20%); max 250%.
- **Compare modes:** 2-up, 4-up, and wipe; sync compare zoom; compare next image.
- **Sorting:** ascending/descending, by EXIF date taken, by star ratings.
- **Find Duplicates:** byte-hash grouping for cleanup.

2.1 Presentation: slideshow & AutoBrowse

- **Slideshow:** 200 transition effects (classic + extended recipes); fullscreen or in-window; folder music via LibVLC; Alt+B / Slides menu.
- **AutoBrowse (signature):** timed multi-panel page turns (Ctrl+A); speed 1–10 seconds; distinct from single-image slideshow.
- **Smart Browse Filter:** filter by rating, keywords, person, favorites, pinned picks — then AutoBrowse the subset.
- **Pin Pick:** shortlist images for Smart Browse from context menu.

Benefit: Review hundreds of shots like a physical light table; run wall displays and kiosk paging without a separate presentation app.

3. Library, session & PostgreSQL catalog

- **Session layer:** unlimited directory slots; hot-reload state.json while running; named Save/Load Settings profiles; archive auto-extract (ZIP/CBZ/7z/RAR).
- **Favorites & Galleries:** session collections; face-driven Person: {name} galleries.
- **Metadata search field:** session filter plus catalog-backed global search when connected.
- **PostgreSQL catalog (optional):** installer can deploy PostgreSQL 18; catalog/sql migrations ship beside the executable.
- **Indexer:** background asset indexing with incremental updates and status-bar progress.
- **Watch folders:** configurable recursive watch with Scan Now.
- **Smart Collections:** parameterized queries — keywords, ratings, path, person, face predicates, camera model, web-only, recently seen, and user presets in smart-collections.json.
- **Photo Exhibitions:** discover museum/gallery shows via web search; browse site-scoped images.
- **Fallback:** app runs fully without PostgreSQL; session JSON remains authoritative for on-screen state.

Benefit: XnView-class DAM when you want it; lightweight folder browsing when you don't.

4. Metadata & IPTC/XMP

- Metadata editor: caption, keywords, dates, extended IPTC fields, star ratings (1–5), EXIF panel.
- GPS coordinates with OpenStreetMap link from context menu.
- Optional writeback to JPEG/TIFF/PNG/RAW via Exiv2 (XMP/IPTC).
- Undo/redo in metadata and editing dialogs.
- Batch Rename with metadata placeholders for plugin actions.
- Cloud-sync conflict detection for OneDrive/Dropbox paths.

5. Face recognition (Phases F1–F5)

Requires PostgreSQL catalog build and Enable face tagging in catalog settings.

Phase	Shipped capability
F1	CPU detection (libfacedetection); Face Tag dialog; IPTC person writeback; library scan
F2	Gallery → People... — browse, Open Photos, Create Gallery, unnamed faces
F3	OpenCV SFace embeddings (ONNX Runtime); auto-suggest; Accept suggestion
F4	Post-index auto-detect on changed paths; Face scan: X/Y status
F5	Bulk match review modal; Find more photos with 3+ training faces

Benefit: Name people once; find them across the library without manual tagging every image.

6. AI, content safety & discovery APIs

- **Photo Chat (AI menu):** multi-turn Q&A about current image; OpenAI, Anthropic, or local Ollama (LLaVA); ai.json configuration; optional Ollama at install.
- **Content APIs:** news, social, and stock feeds as directory slots; Google Image Search; web image search folders (Ctrl+Alt+K).
- **CLIP content safety:** ONNX similarity scoring when models present; heuristic fallback offline; drives Family Safety shields in locked child sessions.
- **CLIP semantic search:** catalog persistence API (pnClipCatalog) — backend module; primary search UI still not a dedicated menu entry in beta.100.

7. Family Safety (child protection)

- Secure-by-default: session locked on every startup until adult PIN entered.
- Multi-adult PIN profiles; salted SHA-256 storage; optional PostgreSQL audit.
- On-image shields in locked mode (not a full-screen gate after certification).
- Blocks Community Live, ChatView outbound actions, and market actions while locked.
- Community → Family Safety... and Settings → General.

Benefit: Shared family PCs stay safe without stripping adult features after unlock.

8. Import, cameras & formats

- **MTP import:** File → Import from Camera... — Windows Portable Devices; connect/disconnect monitor.
- **Camera Manual:** 100-camera catalog; Find Manual Online; WebView2 in-app viewer.
- **Formats:** HEIC, AVIF, ICC color management, ZIP/CBZ/7z/RAR/CBR archives.
- **Camera RAW (LibRaw, beta.99+):** DNG, CR2, CR3, NEF, ARW, ORF, RW2, RAF, PEF, and more; tiered decode — embedded preview / half-size demosaic / full demosaic; WIC fallback.
- **Async thumbnail cache** for responsive grid browsing.

Benefit: Browse and review maker RAW without relying only on Windows codec packs.

9. Editing & batch operations

Tool	Capability
Crop / Resize / Rotate	Crop to view; target W×H; 90°/180°/270°; free-angle bake via PhotoNavigator Kernel (museum sampling + overscan_px margin)
Histogram & Curves	Per-channel curves; brightness/contrast/saturation
Layers	Stack overlays; blend modes; flatten and save
Photo Lab	Computational stack — HDR tonemap, stitch, perspective, illumination
Retouch	Spot heal; clone stamp (Alt+source); brush radius
Mosaic	Grid, hero+L, arc, ellipse layouts; per-tile styling; save as
Filters	Enhance, B/W, Sepia, Mirror; Reduce lossy recompress
Batch	Rename, convert, copy, move, delete; export library/archives

10. Sharing, email & print fulfillment

- **Email:** Gmail API OAuth or default mail-app draft; Email Picture / Email All.
- **Print:** current/all; Windows GDI+ driver dialog.
- **Order Prints:** WHCC Order Submit API and Walgreens Native Photo Prints API v3; prints.json credentials; order history.
- **Archives:** ZIP/CBZ, 7z, RAR/CBR export.
- **Google Drive:** gdrive:// directory slots.

11. Media, voice & accessibility

- **LibVLC 3.0.20:** folder music during slideshow; internet radio streams; catalog persistence.
- **Audio Browse:** Windows SAPI voice commands — Next, Previous, AutoBrowse, Slideshow, Compare, Favorites, Photo Chat, Print, and 30+ command IDs.
- **Toolbar microphone:** Win+H dictation when Audio Browse is off.
- **Remote voice (community):** partner can advance your light table over internet (trust-gated).
- **i18n:** English, Spanish, French, Chinese, Swahili — pnTr() gettext catalogs.

12. PhotoNavigatorGlobalCommunity (shipped phases A–N)

Community-first network layer inside PhotoNavigator — not a separate chat app. Photos, light table, Live video, and capsules are primary; text chat supports them.

Phase	Name	Shipped highlights
A–E	Tree, create, chat, discover	Community hierarchy; ChatView; Discover; Community Live webcam
F	Join & membership	Open join from Discover; membership filters
G	Room delivery	RDS presence heartbeats; multi-peer reachability
H	Connect+	Light-table sync/capture; remote voice commands
I	Programs	Exhibits, competitions, calendar; premiere Live hooks
J	Exchange	.pncapsule manifests; preview-first trade/gift/license
K	Internet scale	Relay directory; WebRTC signaling; live curator badges
L	Trust & identity	Trust tiers; abuse quarantine; phone OTP; federation
M	Market	Rights-first listings; barter/curator/program lanes; print bridge
N	Pen pals	PN-PENPAL invite codes; async capsule gifts to contacts

12.1 Community Live & ChatView (detail)

- **ChatView** (Ctrl+Alt+Shift+M): text, photos, files; LAN TCP port 38472; internet via presence/relay.
- **Community Live**: USB webcam; four share toggles (video, audio, photos, files); WebRTC or LAN fallback.
- **Camera grid in main window (beta.49)**: all room webcams in layout preset grid; double-click zoom any tile.
- **Communities tab**: participant video tiles; room chat; program banners.
- **Photo Capsules**: signed manifests with IPTC/XMP and rights summary; preview-first accept.
- **Market**: barter board, curator shop, program market, verified seller lanes; sponsor badges on Discover.
- **Pen pals (beta.50)**: two-member Personal/Pen pal rooms; Friend trust on accept.

13. Plugins, security & distribution

- **Plugins**: plugins/actions.json — shell commands with path/metadata placeholders; System menu and context menu entries.
- **Code signing**: Azure Artifact Signing for executable and installer (Public Trust path).
- **Auto-update**: GitHub Releases check; downloads installer from PhotoNavigator-Releases.
- **Graceful shutdown**: waits for catalog/community jobs; joins worker threads before SDL teardown.
- **Tests**: 775 Google Test cases; GitHub Actions CI on push to master.

14. Technology stack

Layer	Technology
UI	SDL2, Dear ImGui, Font Awesome 6
Language	C++20, MSVC, CMake
Images	WIC, stb, Exiv2, zlib, LibRaw 0.21
Imaging kernel	pnPhotoNavigatorKernel (C) — filtered rotate + overscan_px
Face / CLIP	libfacedetection, ONNX Runtime

Database	libpq → PostgreSQL 16/17/18
Community RTC	libdatachannel, mbedTLS
Media	LibVLC 3.0
AI	Ollama (LLaVA), OpenAI, Anthropic HTTP
Voice	Windows SAPI
Installer	Inno Setup; Azure code signing; optional PostgreSQL + Ollama

15. Benefits summary

The table below maps major features to the practical benefit for photographers, families, and community operators.

Feature	Benefit
Multi-panel light table (2–36)	Review many photos at once like a physical light table
AutoBrowse + Smart Browse	Hands-free wall displays and filtered kiosk paging
200-effect slideshow	Professional single-image presentations with music
Free rotation + Kernel bake	Fine 0.30° straighten; smooth edges live and on save
LibRaw camera RAW	Review CR2/CR3/NEF/ARW/DNG without extra codec packs
Compare 2-up / 4-up / Wipe	Pick best shot, focus stack, before/after review
PostgreSQL catalog + Smart Collections	Stable DAM index across moves, renames, and PCs
Face F1–F5 pipeline	Name people once; auto-suggest across the library
IPTC/XMP writeback	Metadata survives outside PhotoNavigator
Family Safety + CLIP shields	Child-safe browsing on shared family PCs
Audio Browse voice commands	Navigate while hands are on camera or keyboard-free
Photo Chat (cloud + Ollama)	Natural-language answers about any image
Content APIs + web search	News, social, stock images as browsable folders
MTP camera import	Direct ingest from connected cameras
WHCC + Walgreens print APIs	Order lab prints from the light table
Photo Lab + Curves / Layers / Retouch	Computational and classic edits without leaving the app
ChatView + Community Live grid	Peer messaging and multicam across borders
Photo Capsules + Exchange	Rights-aware trade/gift with preview-first safety
Programs + Market + Pen pals	Exhibits, commerce, and 1:1 global pairing
Trust tiers + abuse quarantine	Safer peer network with policy gates
Plugins (actions.json)	Hook external tools and custom scripts per photo
Signed installer + auto-update	Trusted publisher path; stay current on betas

16. Honest scope notes (beta.100)

- **Live free-rotate matte** — display-only; not written to disk until free-angle bake.
- **CLIP semantic catalog search** — backend + SQL present; no dedicated menu search UI wired yet.
- **Geographic catalog** — latitude/longitude columns and API; no Map View menu in application yet.
- **Market M4–M6** — verified seller storefront UI and open market lanes are partial; core barter/curator/program market shipped.
- **Not Lightroom Develop** — Photo Lab is computational light-edit, not a full parametric develop module.
- **Vision doc remainder** — PHOTONAVIGATORGLOBALCOMMUNITY.md describes future ideas; phases A–N above reflect code shipped today.

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Kirk A Jackson · PhotoNavigator · <https://github.com/jacksonkirka/PhotoNavigator-Releases>